Lesson 1: Length, Distance, and Midpoint of a Line Segment

Counting and subtracting for horizontal and vertical segments, the Pythagorean theorem for slanted ones, and the point exactly halfway.

1. Horizontal and Vertical Line Segments

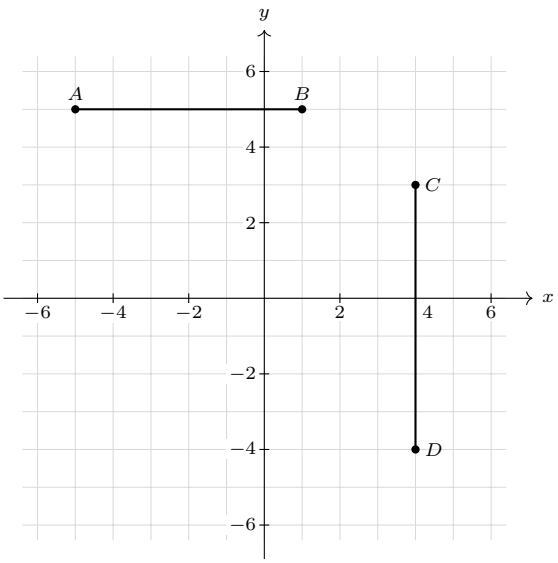
A **line segment** is the piece of a line lying between two points on that line, called its **endpoints**. With endpoints A and B it is written $\overline{AB}$, and its length is written AB. Unlike a full line, a segment stops at each endpoint, so it has a definite length — and a length is never negative.

A segment is **horizontal** when both endpoints share the same y-coordinate, and **vertical** when both share the same x-coordinate. Its length can be found by counting grid squares or, without a grid, by subtracting the coordinates that change and taking the *positive* value. Bars around a number mean "the positive size of": $|a|$ is the size of a with its sign dropped, so $|-9| = 9$ and $|9| = 9$.

$$\text{horizontal: length} = |x_2 - x_1| \qquad \text{vertical: length} = |y_2 - y_1|$$

Example 1 — Horizontal and vertical segments on a grid

Segments AB and CD are graphed below.



(a) Find the length of each segment by counting grid squares.

(b) State the coordinates of the endpoints of each segment.

(c) For AB, subtract the x-coordinates (right minus left). For CD, subtract the y-coordinates (upper minus lower).

(d) Describe a rule for finding the length of a horizontal segment, and of a vertical segment, directly from the coordinates of its endpoints.

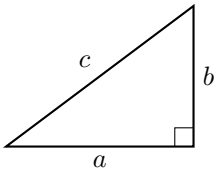
Example 2 — Lengths from the coordinates

(a) Line segment PQ has endpoints $P(5, -2)$ and $Q(-4, -2)$. Determine the length of PQ.

(b) Line segment UV has endpoints $U(3, -2)$ and $V(3, -11)$. Determine the length of UV.

2. Diagonal Segments and the Pythagorean Theorem

Most line segments on a grid run *diagonally*. To find the length of a diagonal segment, build a right triangle under it (or over it — either corner works): a horizontal leg (the **run**), a vertical leg (the **rise**), and the diagonal segment itself as the hypotenuse.



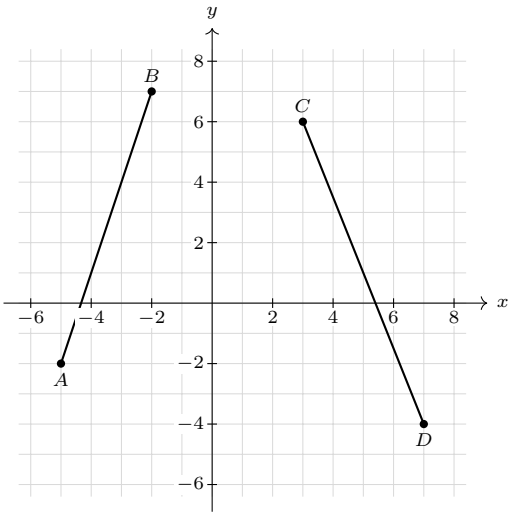
Pythagorean Theorem. In a right triangle, the square of the hypotenuse c equals the sum of the squares of the two legs a and b:

$$a^2 + b^2 = c^2 \qquad \Longrightarrow \qquad c = \sqrt{a^2 + b^2}.$$

On a grid, the run and the rise are the legs a and b, and solving for c gives the segment's length.

Example 3 — Diagonal lengths by the Pythagorean theorem

Determine the length of each segment shown on the grid. Give each answer as **(i)** an exact value in simplest radical form, and **(ii)** a decimal approximation to the nearest hundredth.



(a) Segment AB

(b) Segment CD

3. The Distance Formula

Counting squares stops working when the points are far apart or there is no grid, but the right triangle is always there. For *any* two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ the run is $x_2 - x_1$ and the rise is $y_2 - y_1$, so the Pythagorean theorem always produces the same pattern.

The Distance Formula. The distance d between $P(x_1, y_1)$ and $Q(x_2, y_2)$ is

$$d_{PQ} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Because both differences are *squared*, it does not matter which point is called P and which is called Q: subtracting in the opposite order flips the sign of each difference, and squaring erases the sign either way.

Example 4 — The distance formula

(a) Find the exact length of the line segment from $G(-15, 6)$ to $H(-9, -6)$.

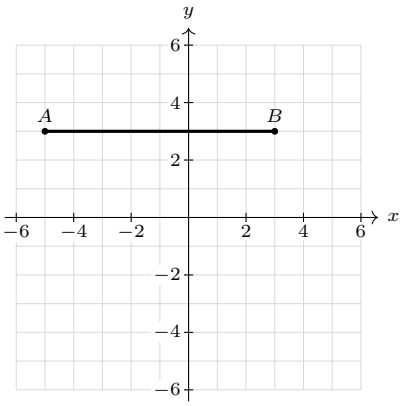
(b) A city places a coordinate grid over a map so that emergency crews can estimate distances. One cell tower is at $(-52, 134)$ and a second is at $(89, 61)$, with each grid unit representing 1 km. Determine the distance between the two towers, to the nearest kilometre.

4. The Midpoint of a Line Segment

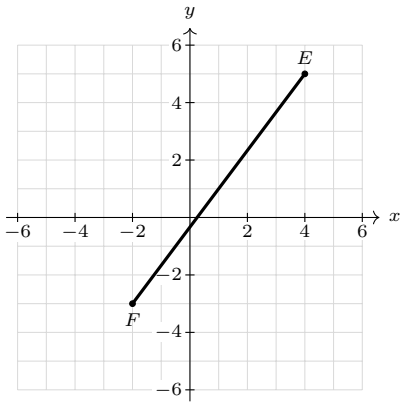
The **midpoint** M of $\overline{PQ}$ is the single point on the segment that is the same distance from P as it is from Q — the point exactly in the centre. It divides $\overline{PQ}$ into two segments, $\overline{PM}$ and $\overline{MQ}$, of equal length. On a grid a midpoint can be found by counting squares: for a horizontal or vertical segment, count in from both ends; for a slanted segment, go *half the run and half the rise*.

Example 5 — Midpoints by counting

(a) Count squares to locate the midpoint M of horizontal segment AB. How do the endpoints' x-coordinates combine to give the x-coordinate of M?



(b) For oblique segment EF, average the x-coordinates and the y-coordinates of E and F *separately* to predict the midpoint M. Then check by counting squares.



5. The Midpoint Formula

The Midpoint Formula. For a line segment joining $P(x_1, y_1)$ and $Q(x_2, y_2)$, the midpoint M has coordinates

$$M\left(\frac{x_1 + x_2}{2},\ \frac{y_1 + y_2}{2}\right).$$

In words: the midpoint's x-coordinate is the *average* of the two endpoints' x-coordinates, and its y-coordinate is the average of the two endpoints' y-coordinates. The distance formula *subtracts* coordinates; the midpoint formula *adds* them and halves.

Example 6 — Midpoints from the formula

Determine the coordinates of the midpoint of the line segment with each given pair of endpoints.

(a) $P(6, 9),\ Q(14, 1)$

(b) $R(-7, 5),\ S(-15, -3)$

6. Working Backward: Finding a Missing Endpoint

Sometimes the midpoint (or part of it) is known and a missing endpoint coordinate has to be recovered. Since

$$x_M = \frac{x_1 + x_2}{2} \qquad \text{and} \qquad y_M = \frac{y_1 + y_2}{2},$$

each coordinate of M supplies an equation that can be solved. Or think of the walk: the midpoint is halfway, so go the same again.

Example 7 — Recovering a blotted coordinate

Devon recorded the coordinates of a trailhead P, a checkpoint M (the midpoint of the trail), and the campsite Q at the far end of the trail. Coffee spilled on the page and blotted out part of the trailhead's coordinates and part of the checkpoint's coordinates:

$$P(x_1,\ 12), \qquad Q(-8, -6), \qquad M(3,\ y_M).$$

(i) Determine the missing x-coordinate of the trailhead, P.

(ii) Determine the missing y-coordinate of the checkpoint, M.

Lesson 2: Rise Over Run: The Slope of a Line Segment

Steepness as a single signed number — counted off a grid, read as a rate of change, and run backwards from a given slope.

1. What Is Slope?

The **slope** of a line segment is a measure of its steepness. It compares the **rise** (the change in vertical position between the two endpoints) to the **run** (the change in horizontal position between the two endpoints):

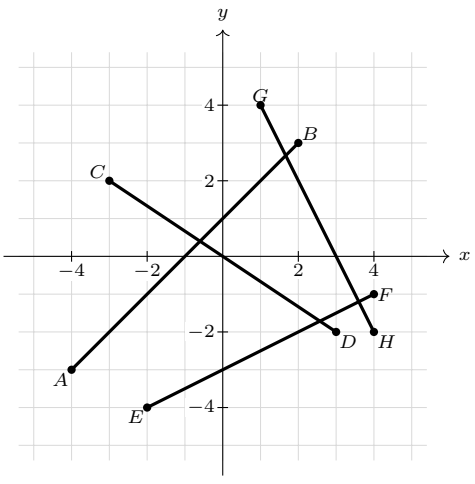
$$\text{slope} = \frac{\text{rise}}{\text{run}}.$$

- The **rise** is POSITIVE if we count UP, and NEGATIVE if we count DOWN.
- The **run** is POSITIVE if we count RIGHT, and NEGATIVE if we count LEFT.

On a grid the rise and run are counted square by square, moving from one endpoint to the other. Counting from left to right keeps the run positive, so the sign of the slope then comes from the rise. A slope that is not a whole number is left as a fraction in lowest terms. (In Lesson 3 this counting is turned into a formula that works straight from the coordinates.)

Example 1 — Rise, run and slope counted off a grid

Each line segment on the grid below has endpoints with integer coordinates. Complete the table.



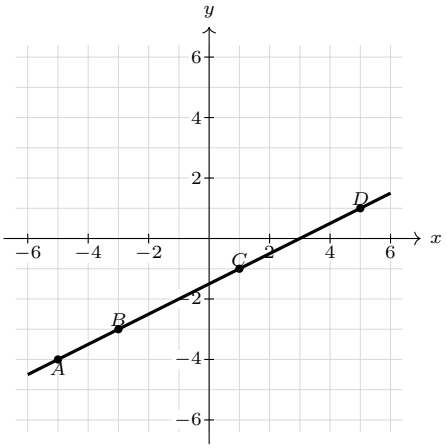
Segment	Rise	Run	Slope
AB			
CD			
EF			
GH			

2. Slope Is the Same Everywhere Along a Line

The slope calculated from *any* two points on a straight line is always the same. That common value is called **the slope of the line**.

Example 2 — The same line, four different pairs of points

Points $A(-5,-4)$, $B(-3,-3)$, $C(1,-1)$ and $D(5,1)$ all lie on the same line. Calculate the slope determined by each pair of points, and state what you notice.



(a) slope of AB

(b) slope of AC

(c) slope of AD

(d) slope of BC

(e) What do you notice?

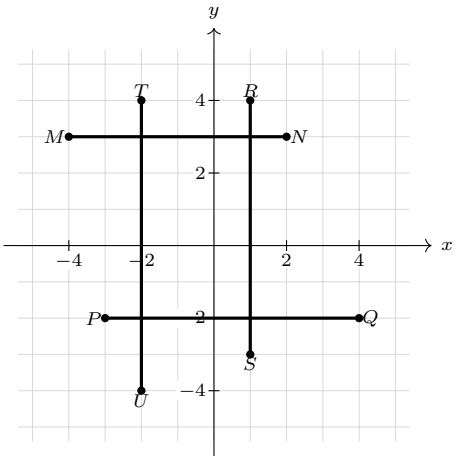
3. Horizontal and Vertical Line Segments

Two special cases deserve their own attention: segments that run straight across (horizontal) and segments that run straight up and down (vertical).

Example 3 — Horizontal and vertical segments

Each segment below has endpoints with integer coordinates.

(a) Calculate the slope of each segment.



slope of $MN =$

slope of $RS =$

slope of $PQ =$

slope of $TU =$

(b) Complete the statements.

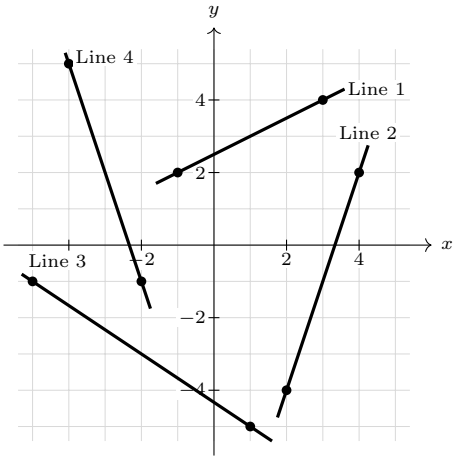
- Horizontal line segments have a slope of _____.
- Vertical line segments have a(n) _____ slope.

4. Positive Slope, Negative Slope

Reading a graph from left to right, a line either climbs, falls, or stays flat. The sign of the slope tells us which.

Example 4 — The sign of the slope

(a) Each line below passes through the two marked points, which have integer coordinates. Calculate the slope of each line.



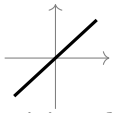
slope of Line 1 = slope of Line 3 =

slope of Line 2 = slope of Line 4 =

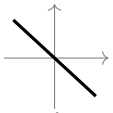
(b) Complete the statements.

- A line which rises from left to right has a _____ slope.
- A line which falls from left to right has a _____ slope.

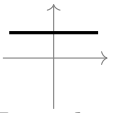
Summary — the four kinds of slope (always read a graph from left to right)



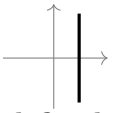
Positive slope
rises from left to right; rise and run have the same sign



Negative slope
falls from left to right; rise and run have opposite signs



Zero slope
horizontal; rise = 0, and $0 \div \text{run} = 0$



Undefined slope
vertical; run = 0, and we cannot divide by zero

5. Steepness: the Size of the Slope

The *sign* of a slope tells the direction of a line; the *size* of the slope tells how steep it is. To compare steepness, ignore the signs and compare the sizes. The size of a slope m is written $|m|$ (read "the absolute value of m"), so $|-5| = 5$ and $|2| = 2$. A line with slope -5 is **steeper** than a line with slope 2, because $5 > 2$ — it simply falls instead of rising.

Example 5 — Ordering lines by steepness

(a) Four lines have slopes -5, 2, $-\frac{1}{2}$ and $\frac{1}{4}$. List the slopes in order from the least steep line to the steepest line.

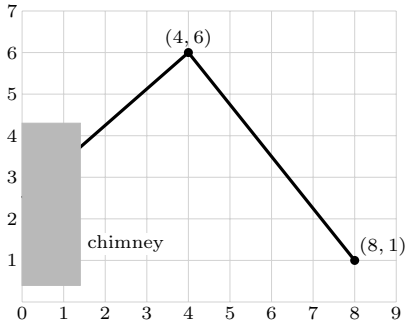
(b) Which line is steeper: one with slope $-\frac{3}{4}$, or one with slope $\frac{2}{3}$? Explain.

6. Slope in the Real World

Slope is how a builder describes a roof, how an engineer describes a road or a ramp, and how a graph shows how fast one quantity changes compared with another. A grid laid over a picture gives a slope estimate, provided two points on the sloped surface can be identified precisely.

Example 6 — The slope of a roofline

A grid has been placed over a sketch of a roof. A chimney blocks part of the roofline to the left of the peak.



(a) Use the grid to determine the slope of the roofline to the right of the peak.

(b) Could this same grid reliably be used to find the slope of the roofline to the left of the peak, behind the chimney? Explain.

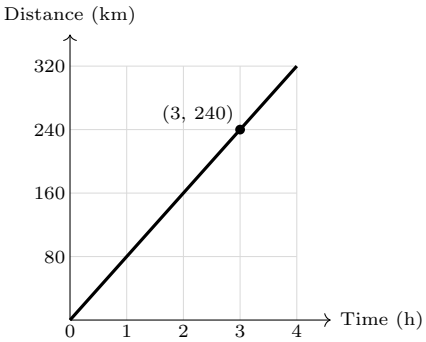
(c) Builders describe a roof's **pitch** as a *positive* rise for every 12 units of run — for example, a "6 in 12" roof rises 6 for every 12 across. State the pitch of the right-hand roofline this way.

Slope as a rate of change. When the axes measure real quantities, the rise and the run have units, so the slope does too: (units on the vertical axis) *per* (unit on the horizontal axis). A slope with units is called a **rate of change**. A positive rate means the quantity is increasing; a negative rate means it is decreasing. Two everyday rates of change:

- A road's **grade** is its slope written as a percent: an 8% grade rises (or falls) 8 m for every 100 m of horizontal run, a slope of $\frac{8}{100}$.
- Building codes require a wheelchair ramp to have a slope of *no more than* $\frac{1}{12}$: at least 12 cm of run for every 1 cm of rise.

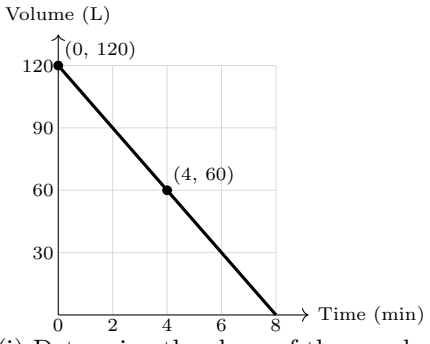
Example 7 — Slope as a rate of change

(a) The graph shows a car's distance from home during a highway trip.



(i) Determine the slope of the graph, including its units.
(ii) What does the slope tell you?

(b) The graph shows the volume of water in a tank as it drains.



(i) Determine the slope of the graph, including its units.
(ii) What does the negative sign mean?

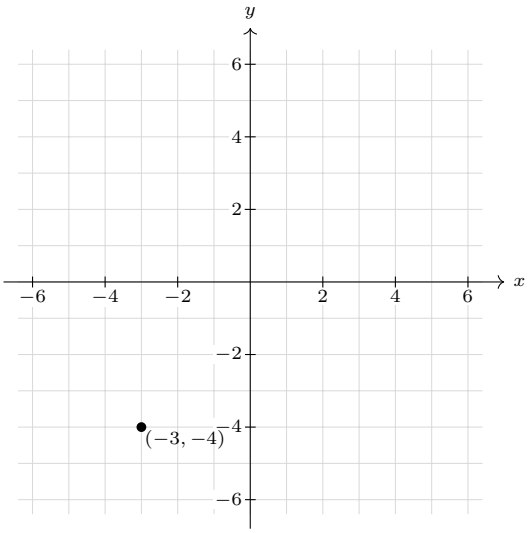
7. Working Backward from a Given Slope

The relationship runs in reverse too. Given a point and a target slope, step off the rise and the run to plot the line; given a slope and one of rise or run, solve for the other.

Example 8 — From a slope back to a line, and to a missing measure

(a) Draw a line segment on the grid which passes through the point $(-3, -4)$ and has a slope of $\frac{3}{2}$. The segment must be long enough to cross both the x-axis and the y-axis. Then write the coordinates of three other points on the segment which have integer coordinates.

Three other points with integer coordinates:



(b) A line segment has a slope of $-\frac{5}{6}$ and a rise of 20. Calculate the run as an exact value.

Lesson 3: Slope Relationships: Collinearity, Parallel Lines, and Perpendicular Lines

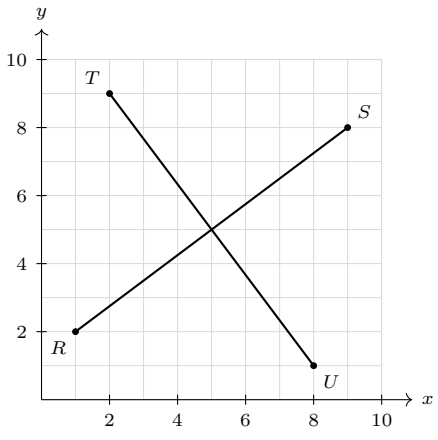
Turning rise and run into a formula — then using slope to test collinearity and to compare two different lines.

1. The Slope Formula

In Lesson 2 slope was found by counting the rise and the run on a grid. Counting needs a graph, and it is slow when the points are far apart. Example 1 shows that the rise is simply a difference of y-coordinates and the run is a difference of x-coordinates — which turns counting into a formula.

Example 1 — Rise over run, and the same slope from the formula

The graph shows line segments RS and TU.



(a) Count grid squares to find the rise and run of RS, then use $\frac{\text{rise}}{\text{run}}$ to calculate its slope.

(b) State the coordinates of R and S. Show that the rise equals $y_S - y_R$ and the run equals $x_S - x_R$, and use these differences to confirm the slope from part (a).

(c) Without counting grid squares, subtract coordinates in the same way to calculate the slope of segment TU.

The Slope Formula. The slope m of the line through $P(x_1, y_1)$ and $Q(x_2, y_2)$ is

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1} \qquad \text{or, with the point names,} \qquad m_{PQ} = \frac{y_Q - y_P}{x_Q - x_P}.$$

Either point may be taken first, provided the coordinates are subtracted in the *same order* in the numerator and the denominator.

Example 2 — The slope formula with negative coordinates

Find the slope of the line which passes through the points $P(-3, 4)$ and $Q(9, -6)$.

$$m_{PQ} = \frac{y_Q - y_P}{x_Q - x_P} =$$

Example 3 — Finding the errors in three students' work

Amara, Ben, and Chloe are each calculating the slope of the line segment with endpoints $W(14, 3)$ and $Z(-6, 11)$.

Amara

$$m_{WZ} = \frac{3 - 11}{-6 - 14}$$
$$= \frac{-8}{-20}$$
$$m_{WZ} = \frac{8}{20} = \frac{2}{5}$$

Ben

$$m_{WZ} = \frac{-6 - 14}{11 - 3}$$
$$= \frac{-20}{8}$$
$$m_{WZ} = -\frac{5}{2}$$

Chloe

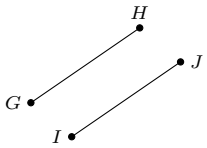
$$m_{WZ} = \frac{11 - 3}{-6 - 14}$$
$$= \frac{8}{-20}$$
$$m_{WZ} = -\frac{2}{5}$$

Since their answers do not all agree, at least two of the three have made an error. Describe every error you can find, and determine the correct slope of WZ.

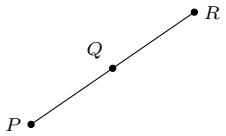
2. Collinear Points

Points are **collinear** if one straight line passes through all of them.

Collinearity test. For three points P, Q and R, calculate m_{PQ} and m_{QR}. If the two slopes are equal *and* the two segments share the point Q, then P, Q and R are collinear. The shared point matters: two parallel segments also have equal slopes, but they lie on two different lines.



$m_{GH} = m_{IJ}$, but no shared point: **parallel**, not collinear



$m_{PQ} = m_{QR}$ and they share Q: **collinear points**

Example 4 — Proving collinearity, and a missing coordinate

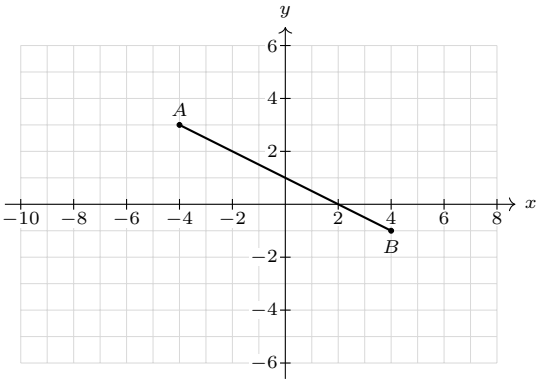
Consider the points $X(-4, 10)$, $Y(2, -2)$ and $Z(11, -20)$.

(a) Prove that X, Y and Z are collinear.

(b) Find the value of w if the point $W(5, w)$ lies on segment XZ.

3. Investigating Parallel Line Segments

Translating a line segment slides every point of it the same distance in the same direction, without turning or resizing it. Because the whole segment moves rigidly, a translated copy always points in exactly the same direction as the original. To translate a point 3 units right and 4 units down, add 3 to its x-coordinate and subtract 4 from its y-coordinate: $(x, y) \to (x + 3,\ y - 4)$. On the grid, segment AB has endpoints $A(-4, 3)$ and $B(4, -1)$.



(a) On the grid, draw the image of AB after each translation, and record the coordinates of the image.

i) 3 units right and 4 units down, forming CD.

$C(___, ___)$ $D(___, ___)$

ii) 5 units left and 2 units up, forming EF.

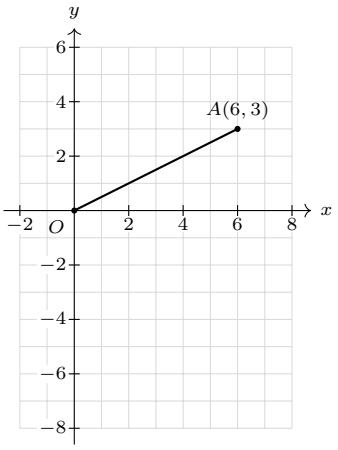
$E(___, ___)$ $F(___, ___)$

(b) Calculate the slope of each of the three line segments.

(c) AB, CD and EF are parallel. Make a conjecture about the slopes of parallel line segments.

4. Investigating Perpendicular Line Segments

Now consider what happens to slope when a segment is rotated 90° about the origin. A rotation of 90° clockwise about the origin sends every point (x, y) to $(y, -x)$: swap the coordinates, then change the sign of the new y-coordinate. For example, $(2, 5) \to (5, -2)$. On the grid, point $A(6, 3)$ has been joined to the origin O to form segment OA.



(a) i) Rotate OA through 90° clockwise about O, draw the image on the grid, and label it OA'. $A'(___, ___)$

ii) Find the slopes of OA and OA', then multiply the two slopes together.

(b) Make a conjecture about the slopes of two perpendicular line segments.

(c) Complete the following to prove your conjecture algebraically. Under a rotation of 90° clockwise about O, point $P(a, b) \to Q(b, -a)$.

$$m_{OP} = \frac{y_P - y_O}{x_P - x_O} = \frac{b - 0}{a - 0} = _____ \qquad m_{OQ} = \frac{y_Q - y_O}{x_Q - x_O} = \frac{-a - 0}{b - 0} = _____$$

$$m_{OP} \times m_{OQ} = _____$$

5. Parallel and Perpendicular Lines

Let two lines have slopes m_1 and m_2.

- The lines are **parallel** if they have the same slope: $m_1 = m_2$.
- The lines are **perpendicular** if the product of their slopes is -1: $m_1 \times m_2 = -1$, equivalently $m_1 = -\dfrac{1}{m_2}$.
- For perpendicular lines each slope is the *negative reciprocal* of the other, provided neither slope is 0.
- **Special case:** a horizontal line (slope 0) and a vertical line (undefined slope) are **perpendicular**, even though their slopes cannot be multiplied. Two vertical lines are parallel.

Line *segments* are parallel (or perpendicular) when the lines containing them are, so the same tests apply to segments.

Example 5 — Reading parallel and perpendicular from slopes alone

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Line segment AC has a slope of $\dfrac{5}{8}$.

(a) Write the slope of line segment GH, which is parallel to AC.

(b) Write the slope of line segment BF, which is perpendicular to AC.

Determine whether each pair of slopes below describes parallel lines, perpendicular lines, or neither.

(c) $m_1 = \dfrac{3}{10},\ m_2 = \dfrac{9}{30}$

(d) $m_1 = \dfrac{4}{9},\ m_2 = -\dfrac{9}{4}$

Example 6 — Slopes parallel and perpendicular to a given segment

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P is the point $(3, 8)$ and Q is the point $(9, -4)$. Find the slope of a line segment

(a) parallel to PQ

(b) perpendicular to PQ

Example 7 — Using slope to find a right angle

$\triangle DEF$ has coordinates $D(-3, 1)$, $E(-1, 5)$ and $F(1, -1)$. Use slopes to show that the triangle is right-angled at D.

Example 8 — Solving for an unknown slope

Two lines have slopes of $-\frac{5}{6}$ and $\frac{k}{9}$ respectively. Find the value of k if the lines are

(a) parallel **(b)** perpendicular

Lesson 4: Characteristics of Linear Graphs

Is it linear? — intercepts, domain and range, how many intercepts a line can have, and matching graphs to their characteristics.

A **linear relation** is a relation whose graph is a straight line. This lesson reads the key characteristics of such a graph — its slope, its intercepts, and its domain and range — and decides from a table of values whether a relation is linear at all.

1. Is It Linear? Tables of Values

On a straight line the slope is the same everywhere (Lesson 2), so equal steps in x always produce equal changes in y. To test a table of values:

- If the x-values go up in **equal steps**, find the change in y from each row to the next. The relation is linear when all of these changes are the same.
- If the x-values do **not** go up in equal steps, divide each change in y by the matching change in x. The relation is linear when this quotient is always the same.

Either way, the constant value $\frac{\text{change in } y}{\text{change in } x}$ is the **slope** of the line — its rate of change.

Example 1 — Linear or not?

Decide whether each table of values represents a linear relation. If it does, state the slope.

(a)

x	0	1	2	3	4
y	-3	-1	1	3	5

(b)

x	0	1	2	3	4
y	1	2	5	10	17

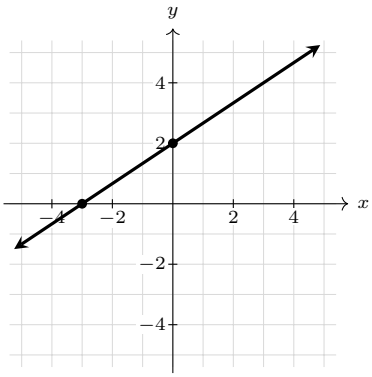
(c)

x	-2	0	3	7
y	11	7	1	-7

2. Intercepts

The **x-intercept** is the x-coordinate of the point where a graph crosses the x-axis; there, $y = 0$. The **y-intercept** is the y-coordinate of the point where a graph crosses the y-axis; there, $x = 0$. An intercept may be reported as a single number or as a point: "x-intercept 4" or "the point $(4, 0)$". In a table of values, look for the row where $y = 0$ (the x-intercept) and the row where $x = 0$ (the y-intercept). In a real situation the intercepts usually have a meaning: the vertical intercept is often the starting value, and the horizontal intercept is often the moment the quantity reaches zero.

Example 2 — Intercepts from a graph and from a table



(a) For the line graphed at the left, state the x-intercept, the y-intercept and the slope.

(b) Show that this table represents a linear relation, then state its intercepts.

x	-4	-2	0	2	4
y	9	6	3	0	-3

3. Domain and Range of a Linear Relation

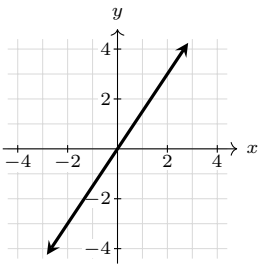
Recall from the relations unit: the **domain** is the set of all x-values of a relation and the **range** is the set of all y-values.

- A line that is neither horizontal nor vertical continues forever in both directions, so every real number is used: domain $\{x \mid x \in \mathbb{R}\}$, range $\{y \mid y \in \mathbb{R}\}$.
- A horizontal line, such as $y = 3$: domain $\{x \mid x \in \mathbb{R}\}$, but the range is the single value $\{3\}$.
- A vertical line, such as $x = -2$: the domain is the single value $\{-2\}$, range $\{y \mid y \in \mathbb{R}\}$.
- A line *segment*, or a line in a real situation, has a **restricted** domain and range, read from its endpoints, e.g. $\{x \mid 0 \leq x \leq 6,\ x \in \mathbb{R}\}$.

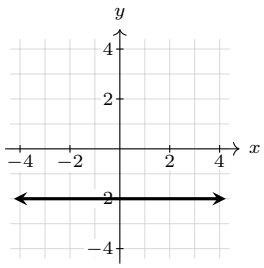
Example 3 — Domain and range of lines and of a real situation

State the domain and range of each line. (The arrows show that each line continues forever.)

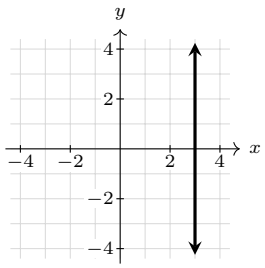
(a)

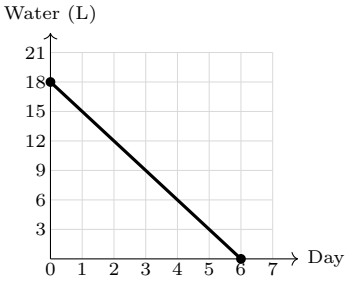


(b)



(c)





(d) An office water cooler starts the week with a full 18 L jug. The graph shows the amount of water left as staff drink it at a steady rate.

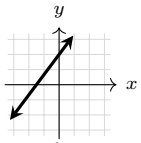
(i) State the two intercepts and explain what each one means.

(ii) Determine the slope, with units, and explain what it means.

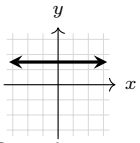
(iii) State the domain and range.

4. How Many Intercepts Can a Line Have?

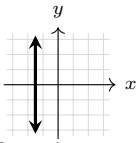
Summary — the number of intercepts of a line



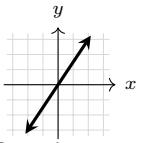
Two intercepts
slanted, not through the origin: one x- and one y-intercept



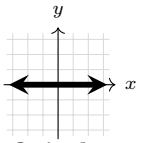
One intercept
horizontal, not the x-axis: a y-intercept only



One intercept
vertical, not the y-axis: an x-intercept only



One intercept
slanted through the origin: both intercepts are 0, one point $(0, 0)$



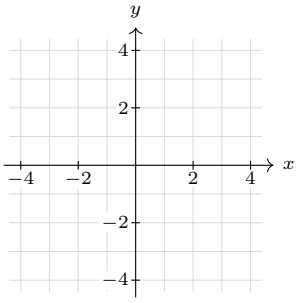
Infinitely many
the x-axis ($y = 0$) or the y-axis ($x = 0$)

Every line crosses at least one axis, so every line has at least one intercept.

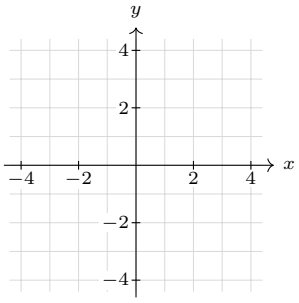
Example 4 — Sketching lines with a given number of intercepts

On each grid, sketch a line with the given characteristics.

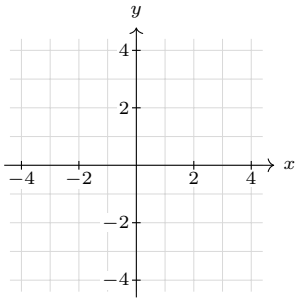
(a) x-intercept 3 and y-intercept -2. State its slope.



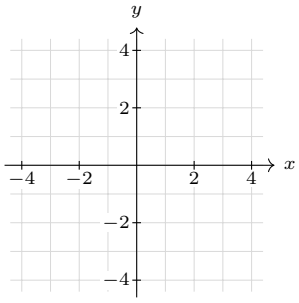
(b) Exactly one intercept, at the point $(0, 4)$.



(c) Exactly one intercept, and a negative slope.



(d) Infinitely many intercepts.

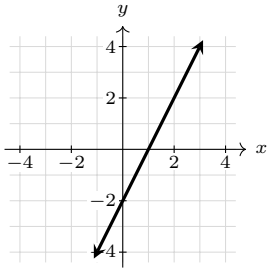


5. Matching Graphs to Their Characteristics

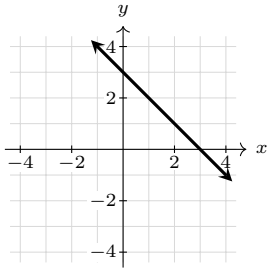
The sign of the slope (rising, falling, flat or vertical) and the intercepts together pin down which line a description fits. Check each characteristic one at a time.

Example 5 — Match each graph with its description

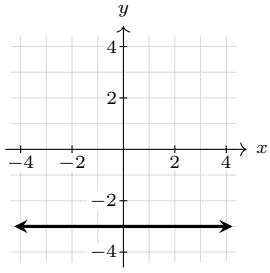
Graph A



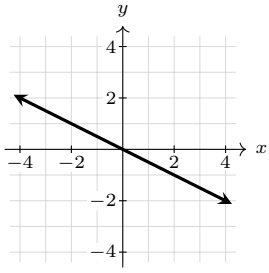
Graph B



Graph C



Graph D



1. Graph _____ Slope 0; y-intercept -3; no x-intercept.

2. Graph _____ Negative slope; both intercepts are 0.

3. Graph _____ Positive slope; x-intercept 1; y-intercept -2.

4. Graph _____ Negative slope; x-intercept 3; y-intercept 3.

Lesson 5: Characteristics of Linear Relations Unit Review

Choosing the right tool — slope, intercepts, domain and range, length and midpoint — and combining them to analyse lines, triangles, quadrilaterals and sets of points.

Sections 1 and 2 review the distance and midpoint formulas from Lesson 1; Sections 3 and 4 review slope and linear graphs from Lessons 2–4; Sections 5–7 combine the tools.

1. The Distance Formula

The distance between $P(x_1, y_1)$ and $Q(x_2, y_2)$ is

$$d_{PQ} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

A **horizontal** segment has equal y-coordinates and a **vertical** segment has equal x-coordinates; for either one the formula collapses to a single subtraction — the positive difference of the two x-coordinates, or of the two y-coordinates.

Example 1 — Lengths of horizontal, vertical and slanted segments

(a) Determine the length of each segment below, and state which one is the longest.
$JK:\ J(3, 10),\ K(9, 10)$ $\quad$ $LM:\ L(5, 13),\ M(5, 3)$
$NP:\ N(-7, 2),\ P(1, 2)$
$QR:\ Q(-4, -3),\ R(-4, -10)$

(b) Determine the exact distance between the points $P(2, 7)$ and $Q(7, -5)$.

2. The Midpoint Formula

For a segment joining $P(x_1, y_1)$ and $Q(x_2, y_2)$, the midpoint is the average of the two x-coordinates paired with the average of the two y-coordinates:

$$M\left(\frac{x_1 + x_2}{2},\ \frac{y_1 + y_2}{2}\right).$$

Run backwards — given the midpoint and one endpoint — each coordinate gives an equation that is solved for the missing endpoint.

Example 2 — The midpoint formula forwards and backwards

(a) Determine the midpoint of the segment joining $C(-6, 3)$ and $D(4, -9)$.

(b) The point $M(2, 5)$ is the midpoint of segment EF. If $E(-3, 1)$, determine the coordinates of F.

3. Slope, Parallel Lines, and Perpendicular Lines

The slope of the segment joining $P(x_1, y_1)$ and $Q(x_2, y_2)$ is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

A rising line has a positive slope and a falling line a negative slope; a horizontal segment has slope 0, and a vertical segment has an **undefined** slope, since its run is 0. The size of the slope, $|m|$, measures **steepness**, and when the axes carry units the slope is a **rate of change** (for example, km/h). Two segments are **parallel** when $m_1 = m_2$, and **perpendicular** when $m_1 \times m_2 = -1$ — that is, when each slope is the *negative reciprocal* of the other. A horizontal line and a vertical line are also perpendicular, even though their slopes cannot be multiplied.

Example 3 — Slope, and the parallel and perpendicular tests

(a) Determine the slope of the segment joining $A(-2, 5)$ and $B(4, -7)$.

(b) Segment GH joins $G(1, 2)$ and $H(5, 10)$. Segment JK joins $J(-3, 4)$ and $K(1, -4)$. Determine whether GH and JK are parallel, perpendicular, or neither.

(c) Determine the slope of a line perpendicular to a segment whose slope is $-\frac{3}{5}$.

4. Characteristics of Linear Graphs

A table of values is **linear** when $\frac{\text{change in } y}{\text{change in } x}$ is constant, and that constant is the slope. The x-intercept is where $y = 0$ and the y-intercept is where $x = 0$. A slanted line has domain $\{x \mid x \in \mathbb{R}\}$ and range $\{y \mid y \in \mathbb{R}\}$; a horizontal line $y = b$ has range $\{b\}$; a vertical line $x = a$ has domain $\{a\}$; a segment or a real situation has a restricted domain and range. A line has one intercept (horizontal, vertical, or through the origin), two intercepts, or infinitely many (an axis).

Example 4 — A linear relation from a table

After a warm spell begins, the depth of snow on a deck, d (cm), is recorded every 2 hours.

t (h)	0	2	4	6
d (cm)	48	42	36	30

(a) Show that the relation is linear, and state the slope with its units.

(b) Determine both intercepts and explain what each means.

(c) State the domain and range of this situation.

(d) The horizontal line through $(5, -2)$ extends forever. State its slope, its intercept(s), and its domain and range.

5. Combining the Tools: Classifying a Quadrilateral

Compute the slope and the length of all four sides. If *both* pairs of opposite sides are parallel and equal in length, the quadrilateral is a **parallelogram**. If in addition one pair of adjacent sides is perpendicular, every angle is a right angle and the parallelogram is a **rectangle**. All four sides equal makes it a **rhombus**, and a rhombus with a right angle is a **square**.

Example 5 — Slope and length of every side

Quadrilateral $WXYZ$ has vertices $W(-5, -1)$, $X(-1, 3)$, $Y(5, 1)$, and $Z(1, -3)$.

(i) Determine the slope and the length of each of the four sides WX, XY, YZ, and ZW.

WX XY YZ ZW

(ii) Use your results from part (i) to determine whether $WXYZ$ is a parallelogram.

(iii) Is $WXYZ$ a rectangle? Justify your answer.

6. Combining the Tools: Testing for Collinearity

Points lying on the same straight line are **collinear**. If P, Q and R are collinear then $m_{PQ} = m_{QR} = m_{PR}$, and showing that any *two* of those slopes are equal is enough. To find which of four points does not belong, compute the slope from one fixed point to each of the other three; the point producing the odd slope is the one off the line. If the fixed point is itself the odd one out, all three slopes will be different — then repeat with a different fixed point.

Example 6 — Finding the point that does not belong

Consider the points $A(-6, -4)$, $B(-2, 2)$, $C(0, 5)$, and $D(4, -8)$. Three of these four points are collinear, and one point does not belong on that line.

(i) Explain, in words, an algebraic strategy for identifying which point is not collinear with the other three.

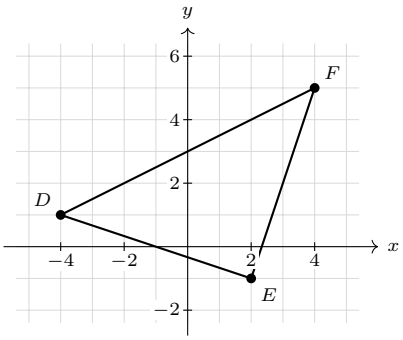
(ii) Carry out your strategy to determine which point is not collinear with the other three.

7. Combining the Tools: Classifying a Triangle

Side lengths classify a triangle: three equal sides is **equilateral**, exactly two equal sides is **isosceles**, and no equal sides is **scalene**. The **converse of the Pythagorean theorem** then tests for a right angle — if the square of the longest side equals the sum of the squares of the other two sides, the angle opposite the longest side is $90°$. The perpendicular slope test $m_1 \times m_2 = -1$ on the two shorter sides gives the same verdict.

Example 7 — Side lengths, and a right angle two ways

Triangle DEF has vertices $D(-4, 1)$, $E(2, -1)$, and $F(4, 5)$, plotted below.



(i) Determine the exact length of each of DE, EF and DF, and classify $\triangle DEF$ by its side lengths.

(ii) Show, in two different ways, that $\triangle DEF$ contains a right angle, and name the vertex at which it occurs.